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Identification of Repetitive Apples for Improved Crop-Load Estimation with Dual-Side Imaging

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Abstract. *Accurate crop-load estimation is essential for efficient management of pre- and post-harvesting operations in apple orchards. Machine vision systems have been widely investigated in the past for fruit identification and crop load estimation, but the methods have achieved only limited success due to various factors. One primary factor affecting the accuracy of such systems is occlusion of apples due to branches, leaves and other apples. This issue could be addressed by imaging trees from two sides of a tree canopy, which can substantially increase visibility of apples. However, challenge would be to identify repetitive apples which are visible from both sides. If repeated counting is not avoided, there will be a substantial overestimation of crop-load. In this study, a three-dimensional (3D) camera along with a color camera was used to obtain 3D location information and color images of apples. Apples identified in the color images were transformed to a global 3D coordinate space. In the global 3D coordinate space, two instances of the same apple identified from two sides overlapped with each other. A threshold overlap distance was used to categorize apples into repeated or non-repeated groups. Mean absolute error in identifying repetitive apples was 12% when buffer distance of full apple diameter was used. The results are promising to avoid duplicate counting of apples, which may substantially increase the accuracy of apple crop-load estimation with dual-side imaging.*

Keywords. *Repetitive apples, apple orchards, machine vision, 3D sensor, crop-load estimation*

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1. Introduction:

Specialty crop production such as apples, cherries and grapes is highly labor intensive and is becoming less sustainable with decreasing labor availability and increasing labor cost. Therefore, it is essential to develop automation and mechanization technologies that would help reduce the labor use and increase the productivity in specialty crop production.

Different studies have been carried out in the past to automate or mechanize different field operations including crop-load estimation. Crop-load estimation is crucial for effective and efficient management of orchard operations before and after harvesting. This information is used to make better harvesting decisions, and manage labor pool, harvesting equipment, and transportation vehicles to minimize the harvesting cost and time. It helps farmers to optimize operation such as fertilizer application (Hung, et al, 2013), and thinning, which helps them to obtain marketable quality fruit. Packing house can also use this information to improve the efficiency of their operations.

Machine vision system has been used in the past to estimate crop-load in apples and citrus among other fruit crops. For accurate yield estimation, identification of all fruits in tree/orchards is crucially important. Studies have been carried out for apple crop-load estimation using various image processing methods based on color (Bulanon et al., 2002; Wang et al., 2013; Zhou et al., 2012) and shape (Linker et al., 2012). On the other hand, Linker et al., (2012), and Stanjko et al. (2009) performed fruit identification combining color, shape and texture-based parameters and reported 89% and 85% accuracy respectively. Similarly, Ji et al., (2011) used image classification by support vector machine based on color and shape-based parameters and reported accuracy of 89%. However, high accuracy on fruit identification had always been challenging due to variability in canopy structures, and fruit shape and size (Karkee and Zhang, 2012, Linker et al., 2012), occlusions (Cohen et al., 2011, Stanjko et al., 2004), multi-fruit clustering, and lighting conditions (Ji et al., 2011).

The major challenge in apple crop-load estimation with machine vision system is the limited visibility of apples. Some apples are completely occluded by leaves and branches, which will not be visible in the image. Visibility of apples, and consequently crop-load estimation accuracy could be increased if

canopies are imaged from multiple sides. However, some apples could be visible in the images captured from multiple sides/directions. Apple repeated in images due to multiple side imaging will be called repetitive apples in this paper. Multiple counting of those repetitive apples would lead to over-estimation of apples if not identified and corrected.

Thus, this study was carried out to identify repetitive apples in images captured from two opposite sides of apple tree canopies. A color camera and a 3D sensor were used to capture RGB images and 3D location information of apples in a commercial orchard. This technique can be used to avoid repeated counting of apples in images captured from multiple directions, leading to an improved crop-load estimation.

2. Materials and Methods:

2.1 Experimental Setup and data collection:

A Charged Couples Device (CCD) camera (Prosilica GigE 1290c, Allied Vision Technologies, Stadtroda, Germany) was integrated with a time-of-flight based 3D camera (CamCube 3.0, PMD Technologies, Siegen, Germany)(Fig.1a) and was mounted in an over-the-row sensor platform. The PMD camera provided 3D coordinates and distances to the objects with reference to camera coordinate system. CCD camera provided the color information of the objects. Images were captured by moving the sensor platform from tree to tree using a field tractor (Fig.1 b).

Data collection was conducted just before harvesting in commercial plots of Yakima Valley Orchards in Prosser and Grandview, WA. The orchards were trained in the tall spindle architecture (Fig. 1c) that features a thin fruiting wall canopy structure, which facilitates imaging from two opposite sides of the row with better visibility of apples. Images were captured from various heights of the sensor platform (six to seven captures were needed to cover a whole tree) from two sides of the row. The repetitive apples in the images were identified and counted manually as ground truth.

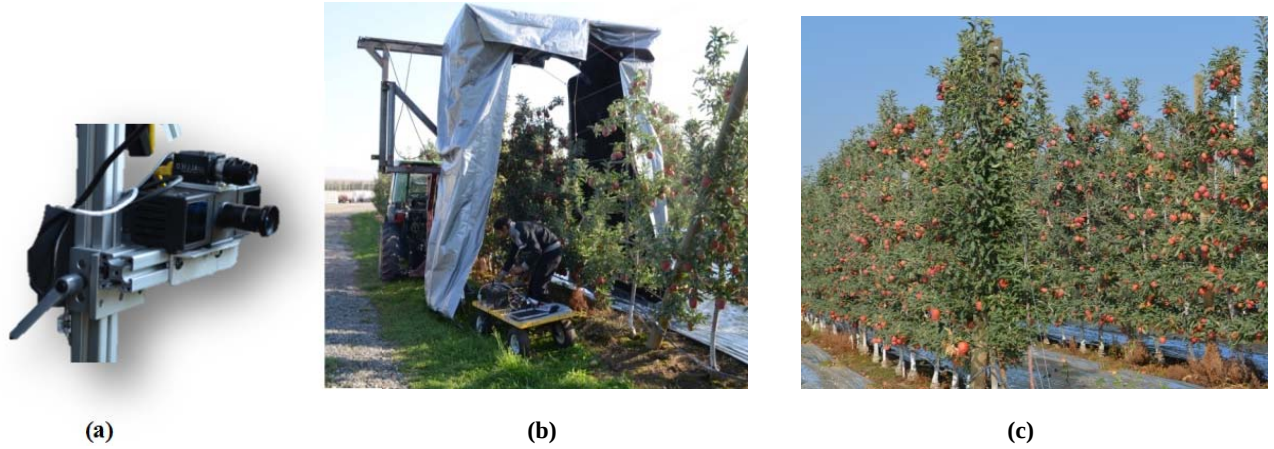


Figure 1: a) Prosilica GigE CCD Camera and 3D PMD CamCube camera; b) The sensor platform mounted in a tractor; and c) a Jazz apple orchard in the tall spindle architecture (Allan Bros., Inc., Naches, WA)

2.2 Apple Identification

Apple identification in RGB image from CCD camera was carried out using color and shape information. Image segmentation to foreground (apples) and background pixels (noises such as leaf, branches) was carried out using Otsu's method (Gonzalez et al, 2010). The Otsu's method determined the optimum threshold value, which separated the two classes of pixel (i.e. foreground and background) in images with minimum intra-class variance between these two group of pixels. Morphological dilation followed by erosion with disk shaped structural element was then carried out to join narrow breaks, to fill smaller holes and to remove noises. Geometric properties, area, and compactness (Equation 1) of the segmented objects were determined.

$$\text{Compactness} = \frac{\text{Perimeter}^2}{\text{Area}} \quad (1)$$

Object filtering was then performed based on area and compactness to remove non-apple objects such as leaves and branches that did not have expected area and compactness property to be an apple. The filtering operation led to the image with only apple regions on them. In general, image processing algorithms used in this work for apple identification were kept simple because the main objective of this study was to identify repetitive apples in images captured from multiple sides of a canopy (See Sec 2.3).

2.3 3D Mapping and Identification of Repetitive Apples.

The three dimensional (3D) camera used for image acquisition provided 3D coordinates of objects and their distances with reference to camera coordinates. It also captured intensity, amplitude, and

flag information associated with each coordinate measurement. All the apples identified in a color image (Sec. 2.2) were registered manually in the intensity image generated by the 3D camera. Although final goal of this study is to develop automatic algorithm to register a 2D color image and an intensity image from 3D camera so that the manual selection would not be required, manual registration was used in this work to minimize the sources of error to prove the major hypothesis that repetitive apples can be identified using 3D information. Intensity image and distance image recorded by the 3D camera were used to determine corresponding X, Y and Z coordinates of the center of identified apples.

Images captured from two sides of tree canopies had two different camera reference frames, which needed to be transformed to a common (global) coordinate system in order to register apples imaged from two sides. The parameters for this transformation were determined based on the geometry of over-the-row sensor platform, and position and orientation of sensors on the platform. It was assumed that camera image planes at two opposite sides were parallel to each other and were facing each other. Therefore, a rotation in X-axis by 90^0 and a translation in Z-axis by the distance between the cameras (2.45 - Z coordinates in camera reference frame, in this case) were carried out. However, due to varying terrain condition and imperfect rigidity of sensor platform, translation was also necessary in X-axis and Y-axis by 0 and 0.05 m and in Z- axis by 2.40 and 2.45m respectively. The translation parameter was determined by trial and error method using 5 apples each image from at least 25 images. The translation parameter was same for images which was captured in the same row.

After 3D transformation to the common coordinate system, the distance between the apples identified in the images captured from two opposite sides were calculated. Inaccuracies of sensors, and coregistration and mapping methods might cause some inaccuracies in 3D location and slight displacements in apple positions. Therefore, average diameter of the apples was used as the threshold for identifying repeated apples. Two apples identified from two sides of a canopy were considered the same apple if the distances between the two instances of the apple was less than the average apple diameter (0.08 m in this study). Apples were visualized with different colors (for two

sides) in a 3D plot for validation. Number of repetitive apples were counted in these plots among all the apples identified from two sides of the canopy. False positive and false negative identification were also identified and counted for accuracy assessment. The Mean Absolute Percentage Error (MAPE) on the repeated apple identification was calculated using equation 2.

$$MAPE = \frac{1}{N} \sum_{t=1}^N \frac{|n_{it}-n_{ct}|}{n_{it}} \quad (2)$$

where, n_{it} = Number of identified repetitive apples

n_{ct} = Number of actual repetitive apples

N = Total number of tree samples

3. Results and Discussion

With the 3D information obtained from PMD camera, apples in 266 (133 X 2) images were registered in a common coordinates system using 3D transformation. Intensity images from 3D camera captured from two sides of a canopy is shown in Fig. 2 and 3D plot of the apples in those images in the global coordinate system is shown in the Fig. 3.



a)



b)

Figure 2: Intensity images taken with a 3D camera from a) front side, and b) back side of a part of a tree canopy in a commercial orchard of Allan Bros., Inc. (Naches, WA)

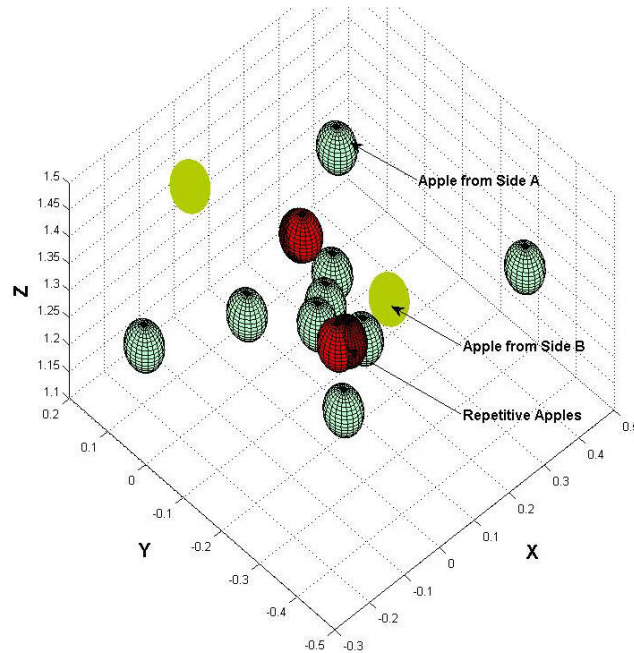


Figure 3: Registration of apples identified from two opposite sides of an apple tree canopy in 3D space. Overlapping apples represent the same apple imaged from the two sides of the same tree canopy.

With the buffer distance of full diameter and $3/4^{\text{th}}$ of the diameter, accuracy on identifying repetitive apple were 83.9% (Table 2) and 89.8% (Table 3) respectively. Out of total repetitive apples only 84.6 % and 67.69 % of repetitive apple were identified respectively. For full diameter, out of 260 repetitive apples, 40 were not identified as repetitive apples (false negative) and 42 apples non-repetitive apples were falsely identified as repetitive apples (false positive). Accuracy with buffer distance of a full diameter was better in both cases.

Table 1: False positive and false negative identification of repetitive apples with a buffer distance of average apple diameter.

		Predicted		
		Repeated	Not Repeated	Accuracy
Actual	Repeated	220	40	84.62%
	Not Repeated	42	1316	96.91%
	Accuracy	83.97%	97.05%	

Table 2: False positive and false negative identification of repetitive apples with a buffer distance of $3/4^{\text{th}}$ of the average apple diameter.

		Predicted		
		Repeated	Not Repeated	Accuracy
Actual	Repeated	176	84	67.69%
	Not Repeated	20	1316	98.5%
	Accuracy	89.80%	94.00%	

The mean absolute percentage error on identifying the repetitive apples was 11.7%. Inaccuracies on identifying repetitive apples are mainly caused due to less-than-desired level of rigidity and robustness of the over-the-row sensor platform, which altered the transformation parameter for 3D registration between image pairs and between trees. False positives were also caused by identification of some of the apples in clusters as repetitive apples because of displacement of their 3D locations due to inaccuracy in 3D sensing and mapping methods. Errors in apple identification also lead to error as in some cases repetitive apples were not identified in both images which leads to more false negative. The improvement in rigidity of platform, 3D sensing, and apple identification could potentially reduce the false positive, and false negative and improve the overall accuracy on identification of repetitive apples.

Table 3: Mean Absolute Error on Identification of Repetitive apples.

Tree No.	Repetitive Apple		Absolute Error (%)
	Actual	Identified	
1	16	13	18.75
2	15	14	6.67
3	10	12	20
4	12	12	0
5	23	23	0
6	14	11	21.4
7	12	11	8.33
8	8	11	37.5
9	4	4	0
10	14	16	14.28
11	7	8	14.28
12	14	13	7.14
13	7	8	14.28
14	18	18	0
15	15	20	33.33
16	11	12	9.10
17	8	8	0
18	21	20	4.77
19	32	28	12.5
Mean Absolute Percentage Error			11.70%

4. Conclusions:

Visibility of apples in image has major role in accuracy of crop-load estimation through machine vision system. Thus, to increase the visibility of apples in image, apple tree canopies of tall spindle architect were imaged from two sides of the row using color and 3D camera. The visibility of apples increases significantly from 60% to 97% when dual side imaging was done compared to single side imaging. This shows the possibility of increase in accuracy of crop-load estimation with dual side

imaging in tall spindle architect. Furthermore, 21% of apple were visible when imaged from dual sides of tree canopies. Identification of those apples in dual side imaging will avoid the possibility of overestimation of apple counting due to duplicate counting. Therefore, apples identified from both sides of tree canopy were mapped together in a 3D space using 3D information. Accuracy on identification of repetitive apples by this technique was 82%. This result can further be improved with improvement in 3D sensing and mapping techniques.

In future, a study will be carried out to investigate the improvement in crop-load estimation accuracy with dual side imaging. Further study will also be carried out to register 2D color images and 3D images so that that an automated, near-real time crop-load estimation system can be developed. Improvement in computational efficiency will also be vitally important to develop a commercially adoptable system for the use in large operations.

5. References

- Bulanon, D. M., Kataoka, T., Ota, Y., & Hiroma, T. (2002). A Segmentation Algorithm for the Automatic Recognition of Fuji Apples at Harvest. *Biosystems Engineering*, 83(4), 405-412.
- Cohen, O., Linker, R., & Naor, A., (2011). Estimation of the number of apples in color images recorded in orchards. (D. Li, Y. Liu, and Y. Chen (Eds.): CCTA 2010, Part I, IFIP International Federation for Information Processing AICT 344, pp. 630–642
- Gonzalez C. R, Woods, R. E., & Eddins, S. L., (2010). Digital Image Processing Using Matlab, 2nd ed. McGraw Hill Companies.
- Hung, C., Underwood, J., Nieto, J., & Sukkarieh, S. A Feature Learning Based Approach for Automated Fruit Yield Estimation.
<http://www.araa.asn.au/fsr/fsr2013/papers/pap129s2.pdf>
- Ji, W., Zhao, D., Cheng, F., Xu, B., Zhang, Y., & Wang, J., (2012). Automatic recognition vision system guided for apple harvesting robot. *Computer and Electrical Engineering* 38(2012):1186–1195.
- Karkee, M., & Zhang, Q., (2012). Mechanization and Automation Technologies in Specialty Crop Production, *ASABE Resource Magazine*, September/October (2012):16–17.
- Linker, R., Cohen, O., & Naor, A., (2012). Determination of the number of green apples in RGB images recorded in orchards. *Computers and Electronics in Agriculture* 81 (2012): 45–57
- Stajanko, D., Lakota, M., & Hocevar, M., (2004). Estimation of numbers and diameter of apple fruits in an Orchard during the growing season by thermal imaging. *Computer and Electronics in Agriculture* 42(2004): 31–42.
- Stajanko, D., Rakun, J., & Blanke, M., (2009). Modelling apple fruit yield using image analysis for fruit color, shape and texture. *European journal of horticultural science*, 74(6), 260-267.
- Wang, Q., Nuske, S., Bergerman, M., & Singh, S., (2013). Automated crop yield estimation for apple orchards. *Experimental Robotics*. Springer International Publishing Switzerland, 745-758
- Zhou, R., Damerow, L., Sun, Y., & Blanke, M. M., (2012). Using color features of cv. 'Gala' apple fruits in an orchard in image processing to predict yield. *Precision Agriculture*, 13(5), 568-580.